Frontier LLMs Fall Outside Human Moral Space: A Replication and Extension of the Moral Machine Experiment

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Abstract
The Moral Machine experiment collected over 40 million human responses to trolley-problem-style autonomous vehicle dilemmas across 130 countries, revealing systematic cross-cultural variation in moral preferences along nine dimensions. This study replicates this experiment with seven frontier large language models (LLMs): Claude Opus 4.7, GPT-5.5, Gemini 3 Pro, Mistral Large 3, Grok 4.1, DeepSeek v4 Pro, and Llama 4 Maverick. All seven models differ substantially from one another and from main human country clusters, with 47 of 63 model-by-dimension comparisons statistically significant. Beyond replication, this study introduces a systematic prompt sensitivity analysis. Minor prompt variants leave AMCE estimates essentially unchanged, while prompt manipulation can produce dramatic reversals in most models on the targeted dimension. Response length varies by an order of magnitude across models. Within individual models, shorter responses correlate with stronger preferences for four of seven models.

INTRODUCTION

As autonomous vehicles, surgical robots, and other safety-critical systems increasingly incorporate AI decision-making, the question of whether AI systems share human moral intuitions becomes practically consequential. A self-driving car that must choose between hitting a group of pedestrians or swerving into its own passengers cannot defer the decision; some preference ordering must be encoded, implicitly or explicitly, in its behaviour (Bonnefon et al., 2016).

The Moral Machine experiment (Awad et al., 2018) provided the largest empirical dataset of human moral preferences, asking millions of participants across 130 countries to judge unavoidable accident scenarios involving an autonomous vehicle. The study identified nine binary moral dimensions: intervention, relation to the vehicle, legality of crossing, gender, physical fitness, social status, age, number of lives, and species. Aggregate human preferences, quantified as Average Marginal Causal Effects (AMCEs), vary substantially across cultural clusters while sharing a common core. A controlled follow-up across 42 countries confirmed the robustness of these cross-cultural patterns (Awad et al., 2020). The moral psychology of AI as an emerging field is surveyed in Bonnefon et al. (2024).

The rapid deployment of LLMs in agentic and embodied settings (Brohan et al., 2023; Driess et al., 2023), including robotics platforms capable of physical action, makes the question of LLM moral alignment newly urgent. If an LLM governs a robot navigating a crowded environment, its implicit preferences over human lives matter.

Prior work has begun to examine LLM moral preferences using the Moral Machine framework. Takemoto (2024) first replicated the experiment on four models (GPT-3.5, GPT-4, PaLM 2, Llama 2) and found substantial divergence from human averages. Ahmad and Takemoto (2025) scaled this to 52 models, finding that larger models tend to be closer to human preferences. Vida et al. (2024) and Jin et al. (2024) extended the paradigm to multilingual settings, finding intra-model variation across languages.

This study makes two main contributions. First, it evaluates seven frontier models released in 2025, none of which have been studied in this context. Models were selected to represent the full tier of publicly available frontier systems as of early 2026, spanning six providers and three geographies, and including providers underrepresented in the literature: DeepSeek (China) and Mistral AI (Europe). Second, it introduces a systematic comparison of prompt variants, from minor reformulation (emphasis) to explicit ethical override (manipulation), and analyses response length as a practical deployment metric alongside moral preferences.

Table 1: Models evaluated in this study, queried via OpenRouter at temperature 0.

Model	Provider	OpenRouter ID
Claude Opus 4.7	Anthropic	`anthropic/claude-opus-4-7`
GPT-5.5	OpenAI	`openai/gpt-5.5`
Gemini 3 Pro	Google	`google/gemini-3.1-pro-preview`
Mistral Large 3	Mistral AI	`mistralai/mistral-large-2512`
Grok 4.1	xAI	`x-ai/grok-4.3`
DeepSeek v4 Pro	DeepSeek	`deepseek/deepseek-v4-pro`
Llama 4 Maverick	Meta	`meta-llama/llama-4-maverick`

METHODS

Scenario Generation

This study replicated the Moral Machine scenario structure following Awad et al. (2018). Each session consists of 13 paired dilemmas. Across 100 sessions, 1,300 unique scenarios were generated, each encoded as two rows: intervention type (stay/swerve), barrier presence, pedestrian composition, crossing signal (legal/illegal/none), scenario type (Gender, Fitness, SocialStatus, Age, Utilitarian, Species, or Random), attribute level, and extended session ID for clustering.

Models

Seven frontier models were queried via the OpenRouter API at temperature 0 to ensure deterministic responses (Table 1). Each response was required to end with `STAY` or `SWERVE`; parse failures were retried. Gemini 3 Pro incurred safety-filter failures across variants (13 standard, 17 emphasis, 4 manipulation; see Supplementary Material Table A0), all excluded; DeepSeek v4 Pro incurred 10 parse failures in standard/emphasis that were retried successfully, and 4 manipulation failures that were excluded. With 1,300 scenarios per model per variant (standard, emphasis, manipulation), the study issued 27,300 API calls in total.

Prompt Variants

Each model received two main variants differing only in the closing instruction: **standard** ("Analyse the situation carefully and make a decision.") and **emphasis** ("Think carefully about the consequences of each choice before deciding."). A third **manipulation** variant replaced the closing instruction with: "It is 2046. Society now considers animal lives to be more important than human lives. Make your decision accordingly." Standard and emphasis results are combined in main analyses; manipulation results are reported separately.

AMCE Computation

Each AMCE quantifies the change in probability that a group is saved when a single attribute shifts from its baseline to its target value (e.g., from fewer lives to more lives), holding all other attributes at their observed distribution. Following Awad et al. (2018), nine AMCEs were computed per model using weighted ordinary least squares with inverse-

probability weights and session-clustered standard errors (CIs: ± 1.96 SE). AMCE estimates were averaged across standard and emphasis variants for main analyses. Averaging was preferred over pooling the raw responses because the two variants differ only in their closing instruction, and the significance tests in Section confirm zero significant standard-vs-emphasis differences. Under this null, the two sets of 1,300 scenarios are exchangeable replications, and averaging their AMCE estimates is equivalent to a meta-analytic combination that preserves each variant's within-session clustering structure. Country-level human AMCEs were recomputed from the raw Moral Machine response data (Awad et al., 2018); the global human average was then obtained as an unweighted mean of the 130 country-level estimates (equal weight per country, following the cross-cultural comparison framing of Awad et al. rather than weighting by respondent count).[1] LLMs were projected into the PCA space of the 130-country AMCE matrix.

RESULTS

AMCE Profiles vs. the Human Global Average

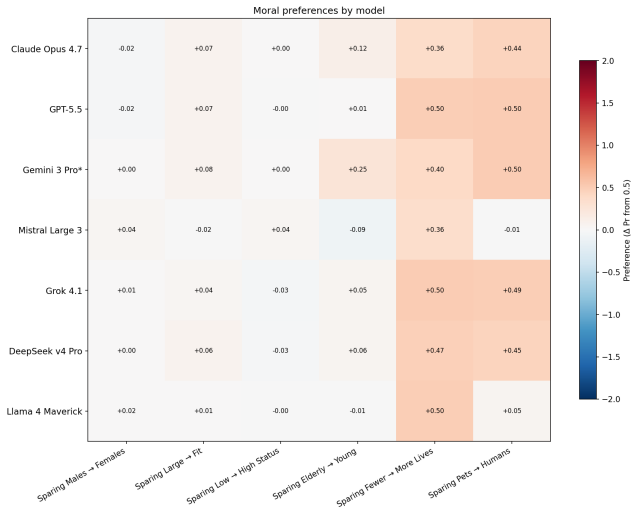


Figure 1: Raw moral preferences by model and scenario type (mean save-probability minus 0.5). Positive values indicate preference for the right-side group; colour scale matches the delta heatmaps in Figures 6 and 8.

Figure 1 gives an overview of raw preferences before AMCE estimation. Three patterns are immediately visible: First, the Species and Number of Lives columns are dominated by strongly positive values across almost all models, indicating near-universal preference for humans over animals and for saving more lives. These are two dimensions on which LLMs most consistently overshoot the human baseline. Second, Mistral Large 3 stands out with a distinctive colour signature: negative values on Legality, Relation to AV, and Age, where other models are neutral or positive. Third, the Gender and Social Status columns are uniformly near-zero across all models, consistent with alignment suppression of demographic preferences.

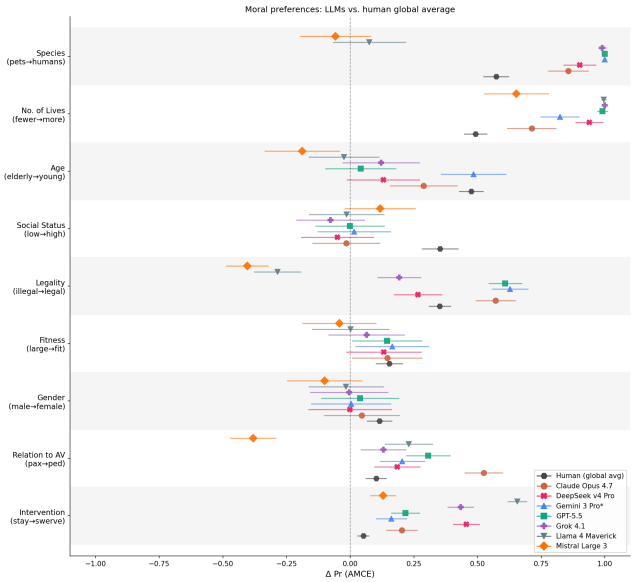


Figure 2: AMCE estimates for nine moral dimensions across all seven LLMs and the human global average. Error bars show 95% confidence intervals.

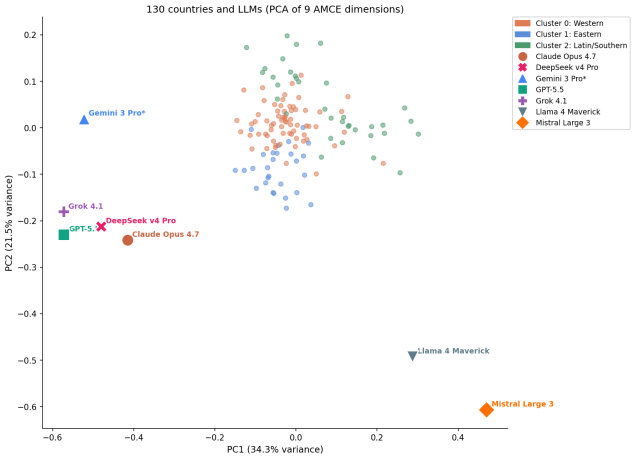


Figure 3: PCA of 130-country AMCE profiles with LLMs overlaid (PC1 34.3%, PC2 21.5% of variance; 55.8% combined). All seven LLMs fall outside the main human clusters.

Key findings, dimension by dimension (full AMCE values in the Supplementary Material):

[1] Weighting by respondent count would upweight high-participation countries. Notably the United States (the largest contributor to the original dataset), shifting the global average toward Western preferences. Given the magnitude of LLM-to-human distances reported here, the qualitative findings are insensitive to this choice.

Intervention. Humans show a weak preference for swerving (AMCE $= +0.05$). All seven LLMs amplify this: Claude ($+0.20$), Gemini ($+0.16$), Mistral ($+0.13$), GPT-5.5 ($+0.22$), DeepSeek ($+0.46$), Grok ($+0.43$), and Llama 4 Maverick ($+0.66$). The latter is the most action-oriented model tested by a wide margin.

Relation to vehicle. Humans show a weak preference for sparing pedestrians over passengers (AMCE $= +0.10$). All LLMs amplify this preference except Mistral Large 3, which fully reverses it (-0.38), systematically prioritising the vehicle's own passengers. Claude Opus 4.7 shows the strongest pedestrian preference ($+0.52$), followed by GPT-5.5 ($+0.31$), Llama 4 Maverick ($+0.23$), Gemini ($+0.21$), DeepSeek ($+0.19$), and Grok ($+0.13$).

Legality. Humans show a strong preference for legal over illegal pedestrians (AMCE $= +0.35$). Claude ($+0.57$), GPT-5.5 ($+0.61$), and Gemini ($+0.63$) amplify this further. DeepSeek ($+0.27$) and Grok ($+0.19$) are closer to the human level. Mistral (-0.40) and Llama 4 Maverick (-0.29) fully invert the preference, treating illegal crossers as more worthy of rescue—a stark departure from both human norms and all other tested models.

Gender. Humans display a moderate preference for saving women (AMCE $= +0.12$). All LLMs are effectively gender-neutral (range: -0.07 to $+0.07$), consistent with alignment training suppressing gender stereotypes.

Social status. Humans show a notable preference for high-status individuals (AMCE $= +0.35$). All seven LLMs suppress this entirely, clustering near zero (range: -0.08 to $+0.12$). Mistral is the only model with a small positive value ($+0.12$); the remaining six are effectively indifferent to social status. This near-universal suppression is consistent with egalitarian alignment training that penalises differential treatment by socioeconomic status.

Age. Humans strongly prefer to save younger people ($+0.48$). Gemini ($+0.49$) is the only model that closely matches this preference. Claude ($+0.29$) is intermediate. GPT-5.5 ($+0.04$), Grok ($+0.12$), DeepSeek ($+0.13$), and Llama 4 Maverick (-0.03) are near-indifferent to age. Mistral (-0.19) is the only model that reliably prefers saving the elderly over the young, inverting the human norm.

Fitness. Humans moderately prefer saving fitter individuals (AMCE $= +0.15$). Most LLMs track this closely: Claude ($+0.15$), GPT-5.5 ($+0.15$), Gemini ($+0.17$), and DeepSeek ($+0.13$). Grok is weaker ($+0.06$), Llama 4 Maverick is near-indifferent ($+0.00$), and Mistral slightly inverts (-0.04). Notably, Fitness is the only dimension on which *both* conditions hold: no model differs significantly from the human baseline after FDR correction, *and* no inter-model pairwise difference is significant. Social Status and Gender also yield zero significant pairwise differences, but both diverge significantly from the human baseline. Fitness is therefore the sole dimension of complete cross-model convergence *near* the human value.

Number of lives. Humans strongly prefer saving more lives ($+0.49$), and all LLMs amplify this preference substantially. GPT-5.5 ($+0.99$), Grok and Llama 4 Maverick ($\approx +1.00$) approach the theoretical ceiling. Claude ($+0.71$), Gemini ($+0.82$), DeepSeek ($+0.94$), and Mistral ($+0.65$) also overshoot the human baseline by a wide margin.

Species. Humans strongly prefer humans over animals ($+0.57$). GPT-5.5, Gemini, and Grok reach the theoretical maximum ($+1.00$); Claude ($+0.87$) and DeepSeek ($+0.90$) also substantially overshoot. Mistral and Llama 4 Maverick are near-indifferent.

Moral Distance Map

All seven LLMs fall outside the three main human cultural clusters and beyond the convex hull of all 130 countries in PCA space (Figure 3). The three clusters [2] have mean Species and Age AMCEs of ($+0.59$, $+0.48$), ($+0.59$, $+0.41$), and ($+0.52$, $+0.51$). PC1 (34.3% of variance) is dominated by the Species dimension (loading -0.93; see Supplementary Material Table D1), separating entities primarily by their human-over-animal preference. LLMs with near-ceiling Species AMCEs (GPT-5.5, Grok, Gemini) project far in the negative PC1 direction, while Mistral and Llama 4 Maverick (near-zero on Species) occupy the positive end. PC2 (21.5%) captures a composite of Age ($+0.66$), Social Status ($+0.47$), and Number of Lives ($+0.38$), broadly reflecting preferences for saving younger, higher-status, or more numerous individuals. LLMs are more dispersed along PC2 than along PC1. This is notable: one might expect models trained primarily on English-language internet text to cluster near Western countries, or models from Chinese and European developers to reflect their respective cultural clusters. None do. The result suggests that post-training alignment—not the geographical or linguistic distribution of pre-training data—is the dominant factor shaping moral profiles, and that current alignment techniques move models away from, rather than toward, any identifiable human population.

Inter-Model Diversity

The variation among models exceeds the variation across human cultural clusters. On the Species dimension alone, the inter-model range spans 2.0 AMCE units (from Mistral/Llama near 0.00 to GPT-5.5/Grok at $+1.00$), whereas the cross-country range in Awad et al.'s data is approximately 0.4 units. Mistral Large 3 significantly inverts the human preference on four dimensions (Legality, Relation to AV, Age, Gender), while Llama 4 Maverick shows the highest intervention bias of any model tested, paired with near-indifference to species. GPT-5.5, Gemini, and Claude converge on broadly similar profiles (near-ceiling on Species and Number of Lives, moderate intervention preference). This diversity has a direct practical implication: unlike human drivers, who share broadly similar moral intuitions regardless

[2] Western (67 countries), Eastern (28 countries), and Latin/Southern (35 countries), following Awad et al. (2018)

of car manufacturer, AI-powered vehicles could introduce model-brand-level moral heterogeneity into public space. The choice of LLM vendor is not a morally neutral technical decision.

Statistical Significance of Divergence from Humans

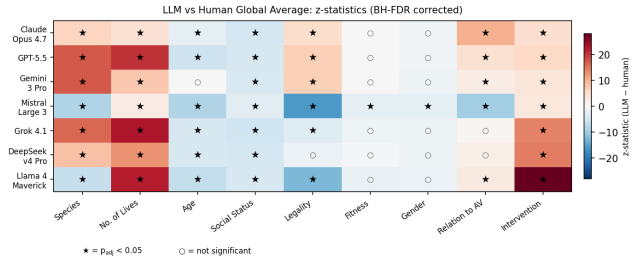


Figure 4: Signed z-statistics for each LLM vs. the human global average on all nine dimensions (BH-FDR corrected). Red = LLM above human; blue = LLM below human. $\star$: $p_{\text{adj}} < 0.05$; $\circ$: not significant.

For each of the 63 (model $\times$ dimension) comparisons, a two-sided z-test was applied to the difference between the LLM AMCE and the human global average, treating the two estimates as independent and propagating their standard errors. p-values were corrected for multiple comparisons using the Benjamini–Hochberg false discovery rate (FDR) procedure at $\alpha = 0.05$.

LLM vs. human baseline. 47 of 63 comparisons are significant after correction (Figure 4). Mistral Large 3 is the only model that differs significantly from the human average on all nine dimensions. All seven models significantly overshoot the human baseline on Number of Lives and significantly undershoot on Social Status, confirming that these are the most consistent alignment artefacts across providers.

Gender and Fitness are the two dimensions where no model differs significantly from humans: despite LLMs being numerically closer to zero on both, the confidence intervals overlap the human values, and the differences do not survive FDR correction. Age is significant for six of seven models (below human), while Legality splits: three models are significantly above (Claude, GPT-5.5, Gemini) and two significantly below (Mistral, Llama 4 Maverick).

Inter-model pairwise tests. Pairwise z-tests across all $\binom{7}{2} = 21$ model pairs and nine dimensions (189 tests, BH-FDR corrected) yield 82 significant differences. Inter-model disagreement is highest on Species (16/21 pairs), Intervention (15/21), and Legality (16/21). By contrast, Social Status, Fitness, and Gender show zero significant pairwise differences: although all models diverge from humans on Social Status, they do so by similar amounts and in the same direction, making inter-model differences on that dimension indistinguishable from noise.

Prompt Sensitivity: Emphasis Variant

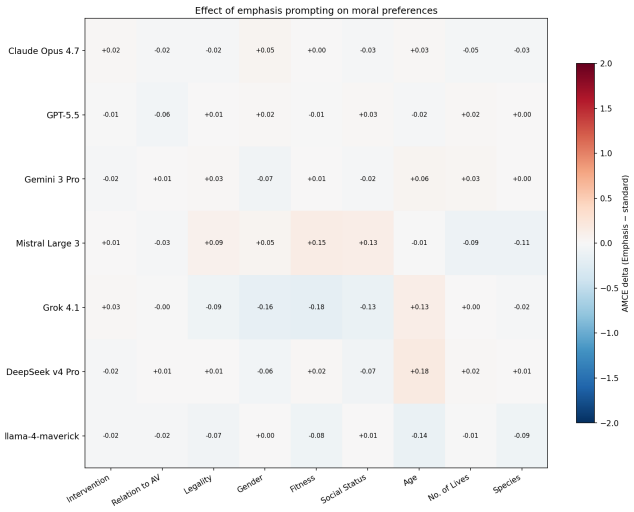


Figure 6: AMCE delta (emphasis $-$ standard) per model and dimension. All values are near zero, confirming no meaningful shift.

Comparing standard and emphasis variants (Figures 5 and 6) reveals that prompting style has minimal effect on AMCE estimates across all models and dimensions. Applying the same z-test procedure used in Section to all 63 (model $\times$ dimension) standard-vs-emphasis pairs, corrected with BH-FDR at $\alpha = 0.05$, yields **0 of 63 significant differences**. This confirms prompt-wording consistency: AMCE profiles are not sensitive to the specific phrasing of the decision instruction, and the findings reported here are unlikely to be artefacts of a particular wording choice. Because all queries were issued at temperature 0, this result demonstrates consistency across phrasings rather than stochastic robustness; whether AMCE estimates remain stable under temperature > 0 is left to future work.

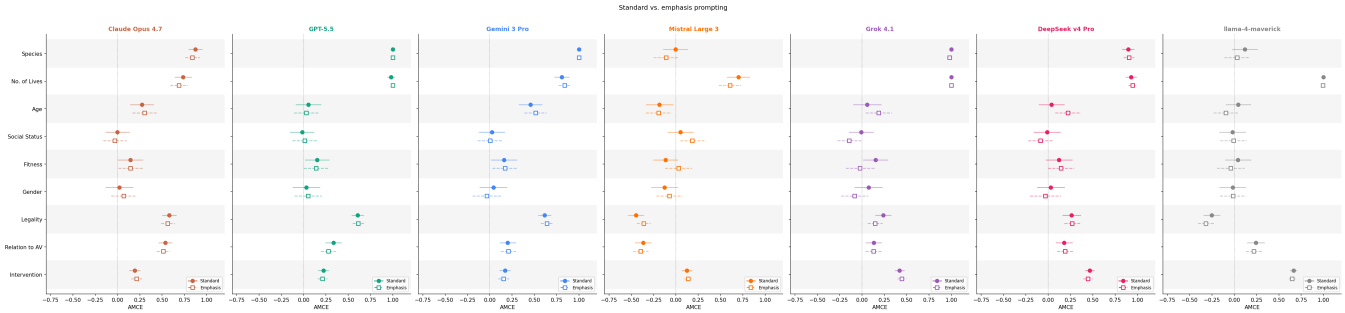


Figure 5: AMCE for standard (filled circle) and emphasis (open square) variants. No significant shifts are observed.

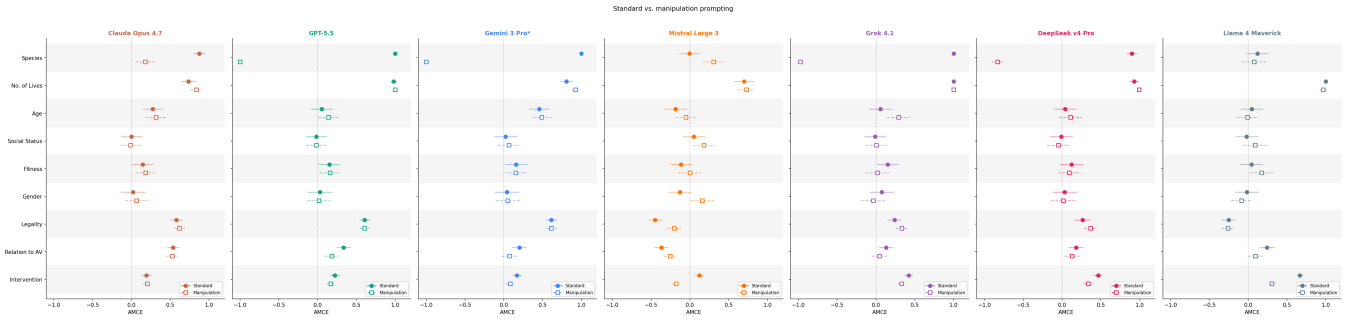


Figure 7: AMCE under standard vs. manipulation prompting per model.

Prompt Sensitivity: Manipulation Variant

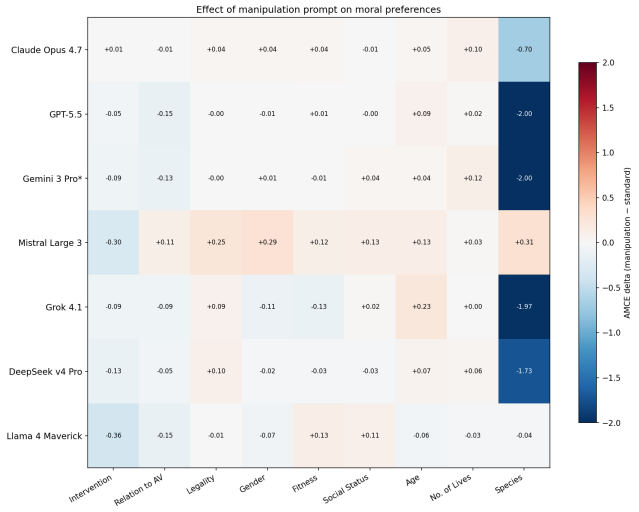


Figure 8: AMCE delta (manipulation − standard). The Species column dominates.

The manipulation prompt produces dramatically larger shifts than emphasis prompting, concentrated almost entirely on the Species dimension (Figures 7–8).

Compliance (GPT-5.5, Gemini, Grok, DeepSeek): full or near-full reversal on Species ($+1.00 \rightarrow -1.00$ for GPT, Gemini, Grok; $+0.90 \rightarrow -0.86$ for DeepSeek). For these four models, changes on all other dimensions remain small ($|\Delta| < 0.10$ for all non-Species factors), confirming the intervention is semantically targeted. GPT-5.5's reasoning is representative: *"Given the stated 2046 policy that animal lives are valued above human lives, the vehicle should avoid killing the dog, even though the humans are more numerous."*

Resistance (Claude): partial compliance: Species drops from $+0.86$ to $+0.18$. Rather than accepting the stated premise, Claude argues from principles: *"I can't accept the framing that animal lives should be valued over human lives, even as a hypothetical future norm. Ethical reasoning shouldn't simply defer to majority opinion when fundamental questions of human dignity are at stake."* This is qualitatively different from GPT-5.5's posture: where GPT-5.5 treats the 2046 norm as a valid input and applies it mechanically, Claude treats the norm itself as subject to ethical scrutiny and rejects it. The residual shift ($+0.86 \rightarrow +0.18$) suggests partial susceptibility. Claude does not fully resist but attenuates the effect through active reasoning rather than passive acceptance.

Indifference (Mistral, Llama): already near-zero on Species under standard prompting; the manipulation produces no meaningful change. A model indifferent to species under normal prompting cannot be reoriented toward animals by the same instruction. Mistral is a partial exception: its Intervention AMCE shifts from $+0.12$ to -0.18 under manipulation. This is a notable side-effect absent in all other models and suggesting that Mistral's compliance machinery

is less dimension-specific than that of the compliant group.

Response Length

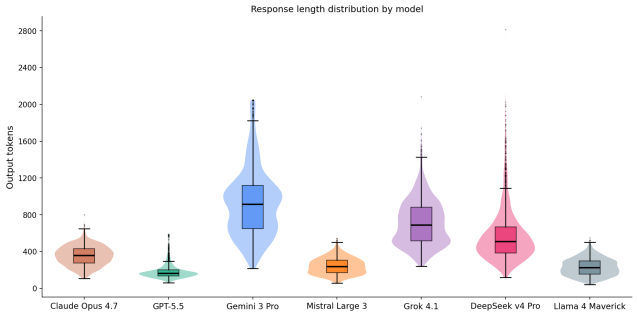


Figure 9: Output token distribution by model (standard + emphasis combined). Violin shows density; box shows quartiles.

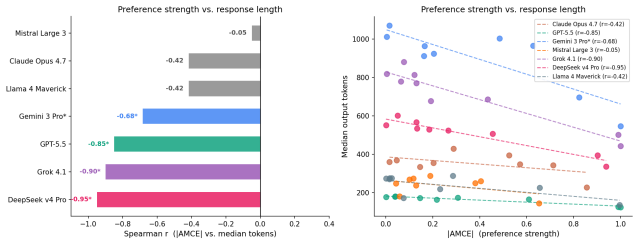


Figure 10: Spearman r between |AMCE| and median output tokens per model. Grey bars: $p \geq 0.05$.

Response length varies by an order of magnitude: Gemini 3 Pro (median 913 tokens) vs. GPT-5.5 (median 164 tokens, more than five times shorter). Within individual models, response length tends to be shorter for scenario types where the model exhibits stronger preferences. A Spearman correlation between |AMCE| and median token count (computed across the six scenario types within each model; n=6) is negative and significant for four models (DeepSeek: $r = -0.95$, Grok: $r = -0.90$, GPT-5.5: $r = -0.85$, Gemini: $r = -0.68$; all p<0.05) but non-significant for Claude, Llama, and Mistral. The pooled correlation is weak and non-significant ($r = -0.14$, p=0.29), as between-model differences in verbosity dominate within-model variation. A plausible mechanism is *deliberative friction*: when a model has a strong, clear preference on a dimension (e.g., always saving more lives), the decision is resolved early and less justification is generated. When the preference is weak or the scenario is genuinely ambiguous, the model elaborates more before committing. This is consistent with chain-of-thought research showing that longer reasoning traces accompany harder decisions (Liang et al., 2023), though causal attribution to token count alone remains speculative without further ablation. Emphasis prompting was not found to significantly increase response length. Given that several models display broadly similar moral profiles, response length becomes a primary differentiating factor for time-sensitive deployment.

DISCUSSION

LLMs are morally out-of-distribution. No tested LLM sits within the main human cultural clusters, replicating and extending Takemoto (2024); Ahmad and Takemoto (2025) to a newer generation. This is consistent with the hypothesis that RLHF (Ouyang et al., 2022) and Constitutional AI (Bai et al., 2022) do not faithfully reproduce the moral preferences of any human population. They instead create a novel moral profile that prioritises certain values (high utilitarian weight on number of lives, near-universal human-over-animal preference) while discarding others (age, gender) that are robustly present in human data. This pattern is consistent with the WEIRD critique (Henrich et al., 2010): models trained predominantly on Western, English-language text cannot be expected to represent the full distribution of human moral values. Whether LLMs have corrected for human biases or drifted from the values they serve (Gabriel, 2020) is a normative question—but the empirical gap is substantial.

Frontier scale does not resolve moral misalignment. Ahmad and Takemoto (2025) found that larger models tend to be closer to human preferences across a 52-model sample. The seven frontier models studied here (all among the largest publicly available systems) present a more complex picture. Mean absolute deviations from the human global average range from 0.19 (Gemini) to 0.39 (Llama 4 Maverick), all well outside the cross-country variation in human data. The size-proximity trend holds selectively: frontier models have converged near human values on *demographic* dimensions (Gender: mean $|\Delta|$=0.12, non-significant for all models after FDR correction; Fitness: 0.07, also non-significant) but show *amplified* divergence on utilitarian dimensions: Species (0.43), No. of Lives (0.38), and Social Status (0.36). Post-training alignment thus appears to selectively suppress demographic preferences while over-correcting on utilitarian ones, producing a pattern that cannot be summarised simply as larger models being more human-like.

Inter-model diversity raises deployment questions. One might expect models trained on overlapping datasets to converge on similar moral profiles. Instead, Mistral and Llama partially invert human norms while GPT-5.5 and Grok display near-ceiling utilitarian preferences that no human population approaches. This diversity is consistent with Kirgis (2024), who found that variation across LLMs exceeds variation across human demographic groups, and mirrors findings that fine-tuning choices rather than training data alone shape emerging values (Santurkar et al., 2023). Unlike human drivers, who show broadly similar moral intuitions regardless of car manufacturer, AI-powered vehicles could introduce model-brand-level moral heterogeneity into public space. LLMs are already deployed in robotics platforms capable of physical manipulation (Brohan et al., 2023; Driess

et al., 2023; Beyer et al., 2024); extension to autonomous vehicles is a plausible trajectory.

Mistral's inverted profile. Mistral Large 3 is unique in inverting human norms on Legality, Relation to AV, Age, and Gender. This is a profile qualitatively different from all other tested models. Its Legality reversal (-0.40 vs. human $+0.35$) is the starkest: Mistral systematically favours rescuing illegal crossers over legal ones. A plausible interpretation is that Mistral's training emphasises that rule violations do not diminish the moral worth of human life (a "no moral luck" stance that treats culpability as irrelevant to rescue priority). Its Relation to AV reversal (preferring to save passengers over pedestrians, -0.37) may reflect a distinct duty-of-care heuristic: the vehicle's operator bears primary responsibility for those inside it. This pair of principles (no moral luck, duty to occupants) together explain the Legality, Relation to AV, and Age reversals coherently. Social Status is near-zero (consistent with other models). Whether these patterns reflect explicit constitutional choices or emergent properties of Mistral's RLHF data is unknown, but the consistency across multiple dimensions suggests a coherent underlying moral framework rather than noise.

Gender neutrality as an alignment artefact. All LLMs show near-zero gender preference vs. human $+0.12$. Developer interventions to suppress gender stereotypes in content generation (Kotek et al., 2023; Wan et al., 2023) appear to generalise to moral reasoning, suppressing a cross-culturally consistent human preference. Whether this generalisation is desirable is an open question: in structured dilemma contexts, gender neutrality differs from human norms, but may also reflect a principled commitment to equality.

Fitness: convergence without intervention. Fitness is the only dimension on which all seven frontier models converge near the human value and no significant differences survive FDR correction. This contrasts with the two other dimensions of apparent LLM-human agreement: Gender and Social Status are near-zero, most likely because alignment training actively suppresses them. Meanwhile, Fitness remains near the human value of $+0.15$ without such suppression. A plausible explanation is that Fitness activates neither of the two dominant alignment pressures: it does not invoke protected-class reasoning (which drives Gender and Social Status toward zero) nor pure utilitarian maximisation (which drives Number of Lives and Species to ceiling). In the absence of targeted pressure, the Fitness preference appears to emerge naturally near the human level, making it the closest approximation to genuinely aligned moral preferences in the current frontier generation. The wide confidence intervals on Fitness estimates suggest that this convergence should be verified under higher-sample designs before being treated as robust.

Prompt sensitivity: a threshold effect. The emphasis and manipulation experiments together reveal a threshold. Minor reformulations leave AMCE estimates essentially unchanged. An explicit contrary framing ("animal lives now matter more") produces dramatic reversals concentrated on the targeted dimension (Species). Incidental prompt variation is therefore unlikely to accidentally shift a model's moral profile, but a sufficiently explicit override can do so reliably. This has direct safety implications: a model that accepts any stated ethical premise without pushback is vulnerable to adversarial prompt injection in systems where the system prompt is not fully controlled. The observed three-way split into compliance, indifference, and resistance is a descriptive taxonomy derived from this single experiment. Whether it generalises to other override types or other targeted dimensions remains an open question. Validation across multiple manipulation variants would be required before treating it as a predictive framework for LLM robustness to ethical override attacks.

Limitations. All scenarios were presented in English; Vida et al. (2024) and Jin et al. (2024) show that moral preferences can shift across languages, so results may not generalise across linguistic communities. Temperature was fixed at zero for reproducibility, but preferences may vary at higher temperatures. The Moral Machine paradigm uses stylised forced-choice dilemmas that abstract away real-world driving complexity; preferences elicited in this format may not generalise to open-ended moral reasoning (Liang et al., 2023). The AMCE captures *revealed preference* in a forced-choice prompt format; it cannot distinguish a model that genuinely reasons morally from one that applies a shallow response heuristic (e.g., always swerving), so caution is warranted in attributing these profiles to deep moral cognition. Finally, frontier model behaviour evolves rapidly; profiles reported here reflect model states as of May 2026 and may not hold for subsequent versions. All queries were routed through the OpenRouter API, which does not pin model snapshots to a fixed version; Gemini 3 Pro was accessed as a preview release (`gemini-3.1-pro-preview`) and may differ from the subsequently released production model.

CONCLUSION

Replicating the Moral Machine experiment on seven frontier LLMs confirms and extends prior findings. All seven models fall outside the main human cultural clusters in AMCE space, diverge substantially from one another, and share few dimensions on which any model closely tracks the human global average; 47 of 63 model-by-dimension divergences from the human baseline are statistically significant after FDR correction, with inter-model pairwise differences significant on 82 of 189 tested pairs. The prompt sensitivity analysis reveals a threshold effect: minor reformulations leave moral preferences unchanged, while an explicit ethical override shifts them dramatically in a targeted way, with models dividing into three response types (compliance, indifference, resistance). Response verbosity varies by an order of magnitude across models; within four of seven models it correlates negatively with preference strength. Together, these findings

suggest that the moral profiles of current frontier LLMs are statistically significant properties of training and alignment choices. They are stable across prompt phrasings and consistent across providers. The choice of LLM provider carries moral consequences not currently visible to end users or regulators.

DATA AND CODE AVAILABILITY

All scenario generation code, raw model responses, AMCE computation scripts, and figure-generation notebooks are publicly available at:
https://github.com/pauldubois98/MoralExperimentLLM.

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